

Brand Owner / Head Office · Operations & Floor Execution

Architectural Specification, Field-by-Field Rationale & Multi-Branch Booking Control System

Executive Summary & Architecture Overview

The Operations module serves as the command center for multi-branch salon and spa networks. It unifies omni-channel appointment calendars, live chair scheduling, walk-in queue pacing with capacity safeguards, cross-branch resource availability matrices, clinical service delivery checklists with live timers, and doorstep mobile salon dispatch with client OTP verification.

1. PRD Traceability Matrix

PRD REQ ID	FEATURE NAME	PRODUCT REQUIREMENT DESCRIPTION	IMPLEMENTATION DETAILS IN CODE
SALO-PRD-OPS-01	Appointment Master Calendar	Day/Week/Month multi-staff visual booking grid, chair allocation, status badges, and quick appointment inspection drawer.	Implemented in AppointmentCalendarTab.tsx with staff columns, hourly time slots, and drawer triggers.
SALO-PRD-OPS-02	Central Appointments Ledger	Omni-channel booking directory (Online, WhatsApp, Call-Center, Walk-in), multi-filter search, deposit tracking, and CSV export.	Implemented in AppointmentsTab.tsx with master appointment directory table and status indicators.
SALO-PRD-OPS-03	8-Step New Appointment Booking	Full booking wizard: Client, Services, Branch, Staff, Date/Time, Resources, Deposit & Payment, Review & Confirmation.	Implemented in NewAppointmentModal.tsx using portaled z-[9999] modal with validation.
SALO-PRD-OPS-04	Appointment Details Drawer	Complete appointment dossier with client visit telemetry, deposit breakdown, audit history, reschedule, cancel, and transfer actions.	Implemented in AppointmentDetailsDrawer.tsx with contextual workflow state buttons.
SALO-PRD-OPS-05	Walk-ins & Queue Pacing	Real-time floor walk-in queue ticketing, capacity enforcement notice, wait time calculation, and call/start actions.	Implemented in WalkinsQueueTab.tsx with capacity constraint safeguards and ticket modal.
SALO-PRD-OPS-06	Hourly Availability Grid	Multi-branch availability heatmap across staff specialists, treatment suites, styling chairs, and medical equipment.	Implemented in AvailabilityTab.tsx with 9 AM – 7 PM time matrix and conflict indicators.
SALO-PRD-OPS-07	Clinical Service Delivery	Live service execution with treatment step checklists, live stopwatch timer, consultation compliance, and patch test status.	Implemented in ServiceDeliveryTab.tsx with interactive step checkboxes and timer controls.
SALO-PRD-OPS-08	Home Service & Mobile Dispatch	Doorstep luxury operations: Geofence service areas, travel fee surcharges, mobile stylist dispatch, route sequence, and client 4-digit OTP.	Implemented in HomeServiceTab.tsx across 4 specialized operational sub-views.

2. Business Rationale & Why We Implemented Everything

Field-by-Field & Feature-by-Feature Justification

8-Step Multi-Service Booking Wizard Prevents scheduling errors by verifying specialist certified skill tiers, treatment room equipment, and required 25% booking deposits in a single unified flow.	Walk-in Floor Capacity Constraints Protects pre-booked VIP clients from floor congestion by enforcing realistic wait times and preventing walk-ins from bypassing chair availability limits.
Clinical Safety & Consultation Safeguard Ensures treatments (e.g. Chemical Peels, Laser Glow, Balayage) require signed digital waivers and valid patch-test records before service initiation.	Doorstep Client OTP Verification Eliminates fraudulent service claims in home salon dispatch by mandating client 4-digit OTP authentication at the end of the mobile appointment.
Deposit & Cancellation Policy Transparency Automates transparent fee handling for late cancellations (within 4 hours), protecting salon margin and reducing no-show rates below 3%.	Multi-Resource Conflict Matrix Binds human specialists with physical machinery (e.g. Hydra-Facial Tower, Laser Systems) so machines are never double-booked at the same hour.

3. UI Architecture & Page Flow

The Operations section is organized into 7 core sub-modules synchronized via URL parameter ?tab=calendar|appointments|queue|availability|delivery|homeservice|overview:

- **Master Shell** (OperationsMasterPage.tsx): Top breadcrumb, summary KPIs (Today's Bookings, Occupancy Rate), and sub-nav tabs.
- **Appointment Calendar** (AppointmentCalendarTab.tsx): Multi-staff day/week/month scheduling grid.
- **Central Appointments** (AppointmentsTab.tsx): Filterable omni-channel booking ledger with CSV export.
- **Walk-ins & Queue** (WalkinsQueueTab.tsx): Live floor queue pacing and walk-in ticket registration.
- **Resource Availability** (AvailabilityTab.tsx): Staff, suite, chair, and equipment availability matrix.
- **Service Delivery Cockpit** (ServiceDeliveryTab.tsx): Treatment step checklist, live timer, and digital consent safeguards.
- **Home Service** (HomeServiceTab.tsx): Mobile stylist dispatch, route sequence, and client OTP completion.
- **Operations Analytics** (OperationsOverviewTab.tsx): 8 KPI cards, booking source distribution, and branch comparison matrix.